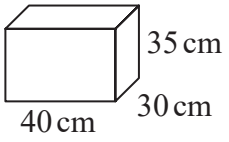
16 Chloe has a van.

She is going to use the van to deliver boxes.
Each box is a cuboid, 40 cm by 30 cm by 35 cm.



The space for boxes in the van has

maximum length	2.4 m
maximum width	1.5 m
maximum height	1.4 m

The space for boxes is empty.
Chloe wants to put as many boxes as possible into the van.

She can put 3 boxes into the van in one minute.
Assume that the space for boxes is in the shape of a cuboid.

(a) Work out how many minutes it should take Chloe to put as many boxes as possible into the van.

Use centimetres: the space is 240 × 150 × 140 cm.

Along the length: 240 ÷ 40 = 6 boxes

Across the width: 150 ÷ 30 = 5 boxes

In the height: 140 ÷ 35 = 4 boxes

6 × 5 × 4 = 120 boxes fill the space exactly.

1 mark
Fit boxes along an edge

1 mark
Find the maximum number

	Boxes	Min	
	3	1	
×40			×40
	120	40	

1 mark
Divide the number of boxes by 3

1 mark
Correct time

40 minutes

(4)

The space for boxes might **not** be in the shape of a cuboid.

(b) Explain how this could affect the time it would take Chloe to put as many boxes as possible into the van.

An irregular space may fit fewer boxes, so loading them could take less time at the same rate.

1 mark
Link fewer boxes to a shorter loading time

(1)

(Total for Question 16 is 5 marks)